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UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 05 – Lecture 02: ML Applications in Banking, Security & Healthcare

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Topic focus: application case studies + risks + evaluation

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.

Quick Links

Banking Security Healthcare Risks Summary

Agenda

Banking

Security

Healthcare

Risks and monitoring

Summary

Learning Outcomes

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- Explain domain-specific risks and why evaluation choices matter.

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- List typical ML use-cases in banking, security, and healthcare.
- Identify common data sources and labels in these domains.
- Explain domain-specific risks and why evaluation choices matter.
- Describe concept drift and why monitoring is needed.

Abbreviations & Keywords

- AML: Anti-Money Laundering
- KYC: Know Your Customer

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- PII: personally identifiable information
- EHR: electronic health record
- Drift: changing patterns over time

KYC: Know Your Customer

PHI: protected health information

Monitoring: ongoing checks

Banking: Typical ML Tasks

- Fraud detection (rare-event classification)

Key issues: imbalance, costs of false positives/negatives, fairness and regulation.

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- Customer support (NLP)

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Banking: Data and Labels (Examples)

- Transaction logs: amount, merchant, time, location, device

Labels may arrive late (fraud confirmed days later), affecting evaluation and monitoring.

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- Transaction logs: amount, merchant, time, location, device
- Customer profile: age, income band, account history
- Labels: "fraud" confirmed, "default" event, churn yes/no

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Security: Typical ML Tasks

- Spam/phishing detection

Key issues: adversarial behavior, evolving threats (drift), false alarm fatigue.

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- Intrusion detection (anomaly detection)
- Malware classification

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Security: What makes it different?

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- The data distribution changes because attackers adapt (adversarial drift).
- FP cost: alert fatigue; FN cost: breach goes undetected.
- Evaluation often needs time-based splits and realistic "red-team" testing.

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Key issues: safety, high cost of false negatives, data privacy, validation and clinical workflow.

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- Imaging assistance (classification/segmentation)
- Triage and decision support

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Healthcare: Data and Workflow Constraints

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- Data: EHR, lab results, imaging, vitals (often missing/noisy)
- Strong privacy and consent constraints (PHI)
- Clinical validation: model must fit workflow and be safe to use

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Cross-Domain Risks (Common Themes)

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- **Bias/fairness**: historical decisions can create unfair models.
- **Privacy**: sensitive data; strict governance required.

Concept Drift (Why Monitoring Matters)

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- Monitoring: track data statistics + performance metrics + alerting.

Monitoring Signals (Examples)

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- Model score distribution (sudden changes can indicate drift)
- Business metrics: chargebacks, false declines, incident volume
- Periodic ground-truth performance once labels arrive

Summary

- Applications differ, but share patterns: classification, regression, anomaly detection.
- Domain risks: imbalance, drift, adversaries, safety and privacy constraints.
- Reliable deployment requires monitoring and careful metric choices.

Sources: NIST, *AI Risk Management Framework (AI RMF 1.0)*, 2023.;Barocas, Hardt, and Narayanan, *Fairness and Machine Learning* (online book), selected sections.;Knaflic, *Storytelling with Data*, selected chapters on communicating results.

Exit Questions

1. Why is class imbalance common in fraud detection?
2. Give one example of concept drift in security.
3. In healthcare, why can false negatives be more costly than false positives?
4. List two monitoring signals you would track after deployment.